



Scan for the package,
the do-file, and this poster

Coding Open-Ended Survey Text Without Leaving Stata: the `catllm` Package for LLM-Assisted Content Coding

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CatLLM

Abstract

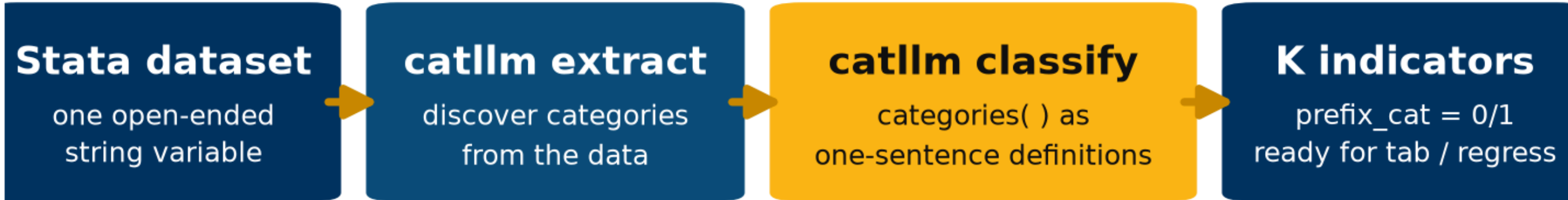
Open-ended survey items capture what closed-ended items cannot, and are routinely dropped from analysis because hand-coding them does not scale. `catllm` is a Stata command that codes free-text responses with large language models and returns ordinary Stata indicator variables. It wraps a Python backend (`cat-stack`) behind five verbs: `extract`/`explore` find categories, `classify` applies them, and `summarize`/`setup` close the loop, across eight providers from OpenAI and Anthropic to fully local Ollama. Tested on multi-label, multi-class items, proprietary models agree with human coders on **97%** of straightforward items and **88–91%** of complex interpretive ones. Models tend to over-classify; three frontier-open models under unanimous voting correct for it, reaching **0.85** precision against Claude Fable 5's **0.78** at little cost in recall: a more balanced classification.

The Problem

- Researchers already use LLMs to code open-ended text, but little guidance exists for keeping classifications aligned with intended meaning without skewing conclusions.
- Model choice is often arbitrary. Researchers overpay for models that excel at general tasks, which rarely predicts coding accuracy.
- Over-classification** is the dominant error: with several categories per response, models tend to assign labels too liberally, especially on ambiguous text. Recall looks fine; precision does not.
- Doing this from Stata usually means exporting to CSV, scripting in Python, and re-merging, which loses the audit trail that makes coding reproducible.
- Aim:** put the whole loop inside Stata, with evidence-based levers on accuracy and over-classification: category definitions, model choice, and cross-provider unanimous consensus.

The Pipeline

Text goes in as a variable — categories come back as variables



*Categories come from extract or your own list;
classify returns multi-label indicator variables either way.*

`generate()` is a prefix, not a variable name: a five-category scheme returns five 0/1 byte variables. `extract`'s specificity, iteration count, and category limit are tunable, so discovery matches the research question instead of a fixed default.

One Command

```
catllm classify response, ///
    categories( ///
        "Housing cost: Moved over rent, a mortgage, or affordability." ///
        "Employment: Moved for a job, a transfer, or school." ///
        "Family: Moved for caregiving or to be near relatives." ///
        "Other: Does not fit any of the above categories.") ///
    apikey($ANTHROPIC_API_KEY) ///
    model("claude-fable-5-1") ///
    generate(reason)

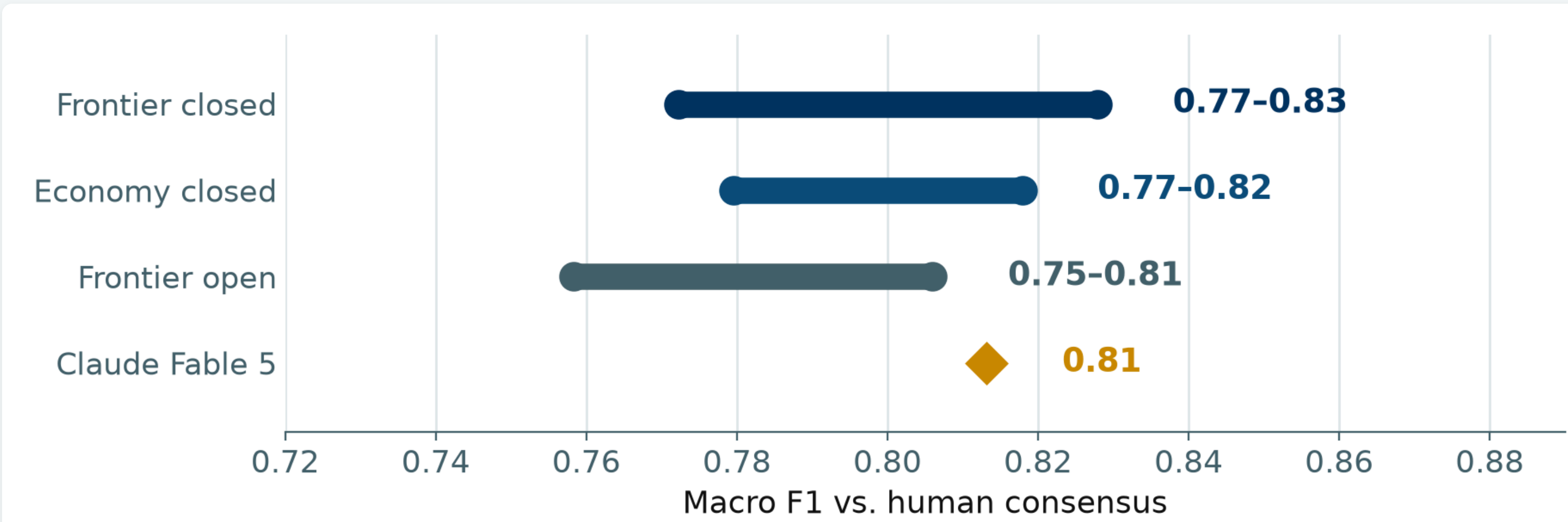
tab1 reason_*
regress reason_Housing_cost age female
```

Each category is a full sentence and `other` is explicit. `catllm` warns on terse labels and offers to add a missing `other` before coding: without a catch-all, the model must guess on responses that fit nothing. `$ANTHROPIC_API_KEY` is a Stata global, set once with `global ANTHROPIC_API_KEY : env ANTHROPIC_API_KEY`.

References

Soria (2026). *CatLLM: A Python package for generating, assigning, and scoring open-ended survey data and images*. | Soria (2026). *High agreement, different stories*. | Soria (2026). *Model diversity over model size*. | Bojic et al. (2025). *Comparing LLM and human annotators*. | Wei et al. (2023). *Chain-of-thought prompting*. | Dhuliawala et al. (2023). *Chain-of-verification*. | Zheng et al. (2024). *Step-back prompting*.

Accuracy Against Human Coders

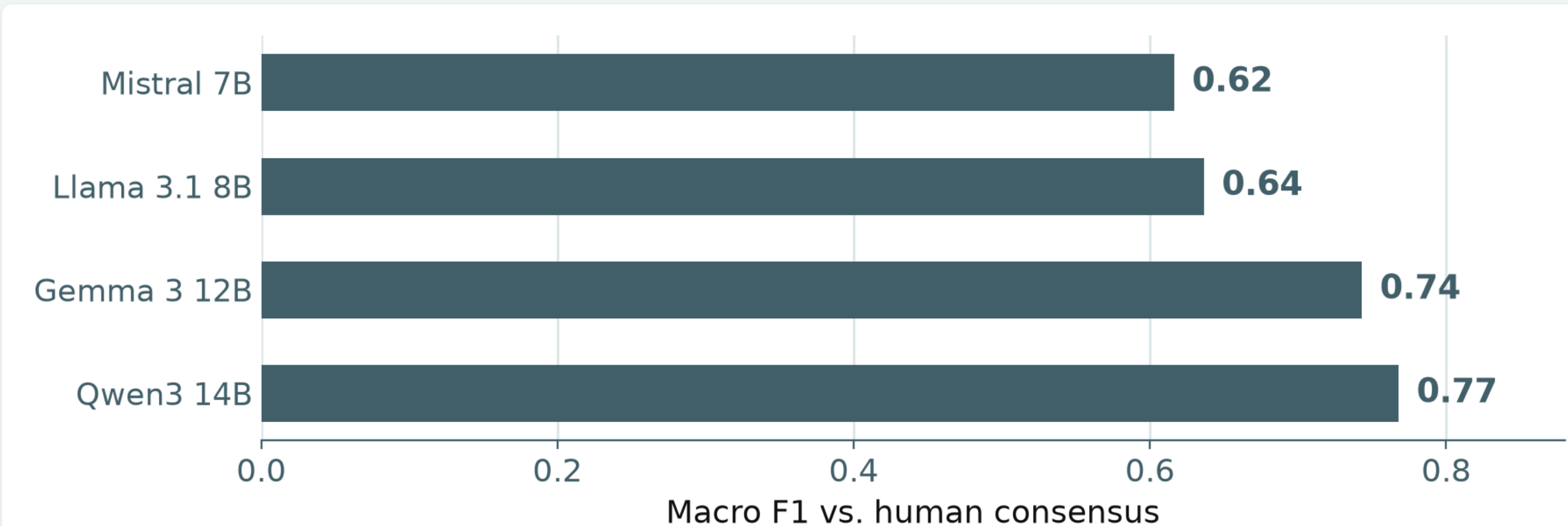


UC Berkeley Social Networks Study validation: macro F1 against human consensus, averaged across three survey questions so no single question dominates. Frontier-closed, economy-closed, and frontier-open tiers overlap closely; Claude Fable 5, shown individually, lands in line with the top cloud tiers. Ranges span the models within each tier.

What Moves Accuracy

- Descriptive categories.** Writing each category as a sentence beats one-word labels on every model tested; `catllm` checks the scheme for terse labels and a missing `other` before any response is coded.
- Ensembles, where they pay.** Unanimous voting over diverse models lifts precision most on interpretive categories and buys little on clear ones; `consensus()` plus per-response agreement scores let you apply it where it helps.
- Minimal prompting by default.** Chain-of-thought, step-back, and expert-persona prompts add tokens without improving survey coding in our tests, so they are opt-in flags, not defaults: cheaper runs, same accuracy.
- Every tier through one command.** Frontier, economy, open-weight, and local models across eight providers are a `model()` argument away, so switching tiers is a parameter change, not a rewrite.

Local Model Spread



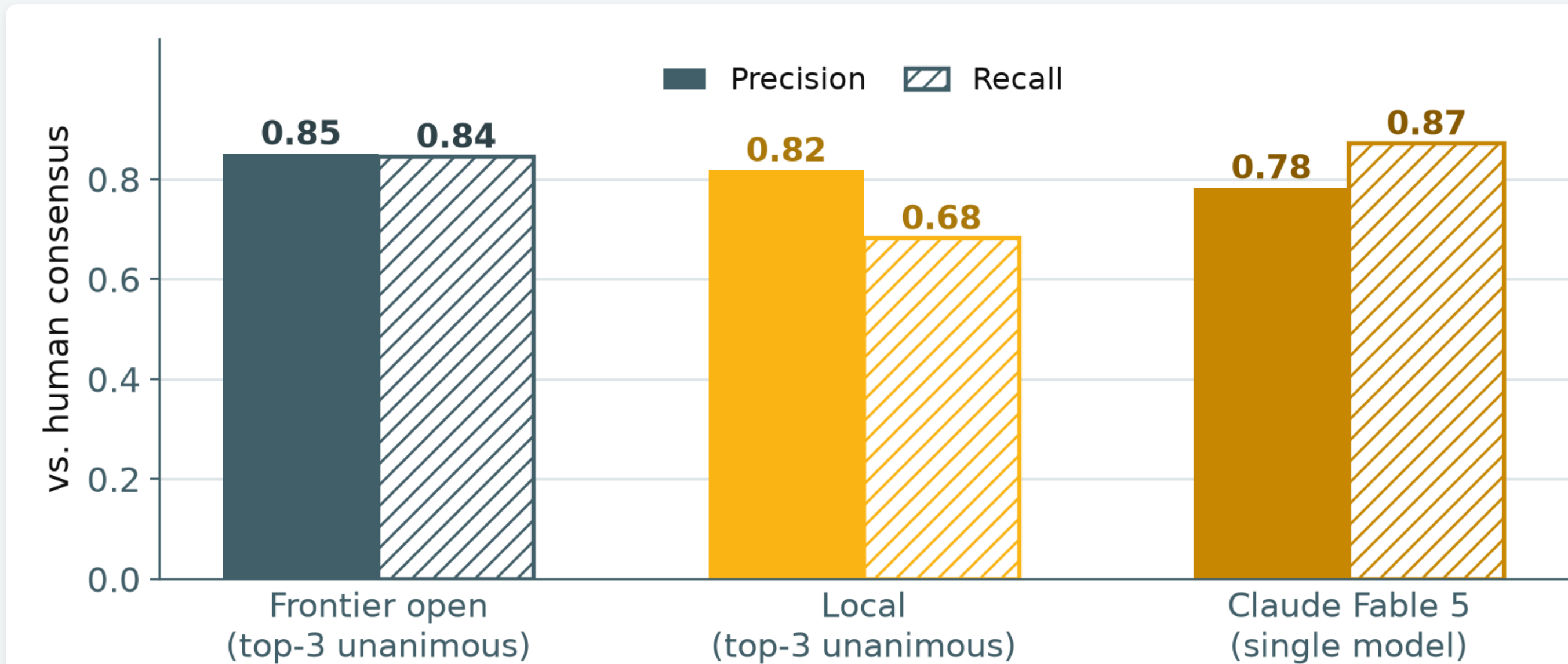
Same macro F1 against human consensus as the cloud tiers, for the four laptop-scale Ollama models. The spread here (0.62–0.77) is about double the cloud tiers' (0.76–0.83): picking well matters far more once you're local.

Choosing a Route

Route	Cost	Speed	Error profile	Data leaves machine
Frontier proprietary	High	Slow	Tends to over-classify	Yes
Small proprietary	Low	Fast	Tends to over-classify	Yes
Multi-vendor unanimous ensemble	Sum of parts	Slowest member	Conservative	Unless all local
Local open-weight (Ollama)	\$0	Hardware-bound	Varies by model	No

Single models of every size tend to assign categories too liberally. A unanimous ensemble of three diverse models corrects for it: pick it when a false positive costs more than a missed code, and when the categories are interpretive, where the precision gain is largest. For restricted-use data under an IRB or data-use agreement, the local route is often the only admissible one, and `catllm` treats it as a first-class route: any Ollama model goes in `model()`, the server starts on demand, and JSON recovery covers the smaller models.

Unanimous Ensembles



`consensus("unanimous")` assigns a category only if all models agree. Fable 5 assigns 38% more labels than coders on the hardest question. The three best frontier-open models beat it on precision (0.85 vs. 0.78) and F1 (0.85 vs. 0.82) for 3 points of recall: a more balanced coder. The best local trio matches that precision and reaches 0.74 F1 at \$0, 8 behind Fable 5; weak members cost recall.

+38%

extra labels on the hardest
question, Claude Fable 5

0.85

precision, top-3 frontier-open
unanimous ensemble

\$0

fully local
route

Key Findings

• 1 command

Code text without leaving Stata. `catllm classify` turns a string variable into 0/1 indicators ready for `tab` and `regress`; `extract` discovers the categories first, so the codebook comes from the data, not a guess

• 0.85 vs. 0.78

Run an ensemble with one option. `consensus("unanimous")` over three diverse models reaches 0.85 precision against 0.78 for Claude Fable 5 and beats it on F1, correcting for over-classification with no hand-merging

• 8 providers

Choose the route your data allows. One command runs on OpenAI, Anthropic, Google, Mistral, xAI, HuggingFace, Perplexity, or fully local Ollama at \$0, where the package auto-starts the server and recovers JSON from small models

Reproduce This

Everything on this poster runs from one self-contained do-file: no data downloads, no file paths, guarded so it completes with a single API key or with none at all.

```
net install catllm, ///
    from("https://raw.githubusercontent.com/chrissoria/cat-llm/main/stata-package/") ///
    replace

catllm setup
do catllm_stata_conference.do
```

The file walks the full pipeline: classify, analyze the resulting indicators, compare three providers, run an ensemble with agreement scores, discover a coding scheme with `extract`, and repeat the whole thing locally at zero cost. Nine further worked examples ship with the package.

Limitations

- Concordance with human coders is not validity.** Where humans systematically misread a response, an agreeing model does too; agreement measures replication of a benchmark, not correctness.
- Validation is strongest on English-language survey text** of the kind studied here. Performance on other languages, registers, and domains needs separate checking on your own data.
- Unanimity trades recall for precision.** On the most interpretive question the top-3 ensemble gives up 11 points of recall against Fable 5 to gain 11 of precision; weak local members triple that cost. Choose by what a false positive versus a missed code costs you.
- The ensemble evidence here is small.** Three questions at 175 responses each, one interpretive, and the trio was picked on the data it is scored on (17 of 20 trios still beat Fable 5 on F1). The pattern replicates on 3,208 responses and 16 earlier models in Soria (2026), *Model diversity*.